

Assignment 4

Object-Oriented Programming in Java — Using ArrayList

Total Marks = 20 (Question 1 [Part A = 10, Part B = 10])

General Instructions

All parts must be implemented using **java.util.ArrayList** to store objects (do not use plain arrays). Question 1 is split into **Part A** and **Part B** so it can be completed in stages; Part B continues directly from the classes and ArrayLists built in Part A. Question 2 is independent of Question 1. Each part/question must be submitted as a separate, compilable Java program. Follow proper encapsulation (private fields with getters/setters where needed), meaningful variable names, and clean, well-indented code.

Part A

You are required to design a Java program for a **Library Management System**. The system should manage **Librarians** and the **Books** assigned to them, using an **ArrayList** to store each collection of objects. Part A covers the class design, object creation, and the basic Add / Display operations. (Delete and Update operations are covered separately in Part B.)

[Part A Marks = Object Creation = 5 + Menu Creation (Add & Display) = 3 + Clean Coding = 2 = 10]

Librarian

```
- name : String
- librarianId : String
<<constructor>>Librarian()
+ toString() : String (returns librarian details in a readable format)
```

Book

```
- title : String
- bookCode : int
- price : double
- librarian : Librarian
- category : String (Fiction, NonFiction, Reference, Periodical)
- libraryCode : int (static variable, common for all the books)
<<constructor>>Book()
<<constructor>>Book(Librarian) (creates a book with an assigned librarian)
+ updateBookRecord() : void - updates details such as price or category (implemented but
not yet wired into a menu option - that comes in Part B).
+ getBookDetails() : void - displays complete book info.
+ getLibrarian() : Librarian - returns the librarian assigned to the book.
```

The Java program should store all Librarian objects in an **ArrayList<Librarian>** and all Book objects in an **ArrayList<Book>**. First, create Librarian objects and then create Books. The maximum allowable limit for Librarians and Books is **2** and **5** respectively (checked using the ArrayList's size). If this rule is violated, an error message should be displayed and the object should not be created/added to the ArrayList.

For Part A, the menu should have the following options only:

- Add Librarian
- Add Book
- Display Book details
- Display Librarian details (of a particular book)

Part B

Continue your solution from **Part A**. Extend the same **Librarian** and **Book** classes and the same **ArrayList<Librarian>** / **ArrayList<Book>** collections to add full Delete and Update support, so the program now offers the complete menu below.

[Part B Marks = Menu Creation (Delete & Update) = 5 + Test Cases Success = 5 = 10]

Deleting a Librarian or Book should remove it from the corresponding ArrayList. Updating a Book should call `updateBookRecord()` to change its price/category; updating a Librarian should change its name. Re-test the maximum limit rule (2 Librarians, 5 Books) after your Add operations, and re-test Display after your Delete/Update operations to confirm the ArrayLists reflect the changes correctly.

The complete menu for Part B should have the following options:

- Add Book
- Delete Book
- Add Librarian
- Delete Librarian
- Update Book
- Update Librarian
- Display Book details
- Display Librarian details (of a particular book)